What the Leftover Hash Lemma Gives

And What It Cannot: a Limitation on the Collision Route

Resolves. Nothing. This is a companion note to *Randomness Extraction from Unpredictable Random-Oracle Sources: A Leftover-Hash-Lemma Conjecture, Public Seed* (`main.tex` in this folder), whose conjecture is proved in `proof.tex` beside it. Two things are recorded. On a fixed universal table the leftover hash lemma returns the conjectured bound's first summand exactly (Proposition 2). No table available at the seed lengths at issue is universal, and on the tables that are available the lemma returns *at least* $\frac{1}{2}\sqrt{(W-1)/K}$ for a source it cannot exclude, where W is defined in Theorem 1 and is of order $\min\{R, D\epsilon\}$; under the side condition of Lemma 1 that floor is a statement about the conjecture's *second* summand, where it costs $\log_2 W$ bits of seed against the conjectured $\log_2\log_2 D$. Section 8 records that the resulting bound is *not new* — it is the seed-length limitation of the leftover hash lemma, known since Stinson and stated as such by Barak et al. [1] — and says what the note adds instead. Section 10 says what is *not* claimed, and in particular that no matching upper bound is proved: the floor is one-sided.

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1 Setting

The setting is that of `main.tex`, repeated here so this note stands alone. Fix nonempty finite sets $\mathcal{K}$ (seeds), $\mathcal{D}$ (inputs), $\mathcal{R}$ (outputs), and write $K := |\mathcal{K}|$, $D := |\mathcal{D}|$, $R := |\mathcal{R}|$. A table H is drawn uniformly from $\mathsf{Fun}(\mathcal{K}\times\mathcal{D},\mathcal{R})$. A source S, given the *entire* table, outputs a pair $(x,z)\in\mathcal{D}\times\{0,1\}^*$; a seed $sd \xleftarrow{\$} \mathcal{K}$ is drawn *after* S has terminated; and an unbounded distinguisher D, given (H, sd, y_b, z), is to tell $y_0 = H(sd,x)$ from a uniform y_1. Write $\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D})$ for twice its winning probability minus one. The source is *ϵ-unpredictable* if no unbounded predictor given (H,z) guesses x with probability more than ϵ; equivalently, if $\mathbb{E}_{(H,Z)}[\epsilon_{H,Z}]\le\epsilon$ in the notation below. Conjecture 1 of `main.tex` asserts a universal constant c such that, for every ϵ-unpredictable S and every unbounded D,

$$\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D}) \;\le\; c\left(\sqrt{\epsilon R}+\sqrt{\frac{\log_2 D}{K}}\right). \tag{1}$$

Every numerical comparison below uses (1) exactly as displayed, with $c=2$, the constant `proof.tex` establishes. No other constant from that note is imported and none is needed.

Throughout, condition on a table and an auxiliary output, $(H,Z)=(h,z)$, and write $p_x := \Pr[X=x \mid h,z]$ for the source's conditional law and $\epsilon_{h,z} := \max_x p_x$. The quantities

$$\mathrm{CP} := \sum_{x\in\mathcal{D}} p_x^2, \qquad c_k := \Pr_{X,X'}\big[h(k,X) = h(k,X')\big], \qquad \bar{c} := \frac{1}{K}\sum_{k\in\mathcal{K}} c_k,$$

all depend on (h,z), which the notation suppresses; where they appear inside an expectation over (H,Z) they are to be read as the corresponding random variables. Here X,X' are independent draws from p. Note $\mathrm{CP} \le \epsilon_{h,z}$, since $\sum_x p_x^2 \le (\max_x p_x)\sum_x p_x$.

Fact 1 (Cauchy–Schwarz, two forms). Let q be a probability distribution on a finite set Ω and let $A \subseteq \Omega$ carry all of its mass. Then

$$\sum_{\omega} q_\omega^2 \ \ge\ \frac{1}{|A|}. \tag{2}$$

If moreover u is uniform on Ω, then

$$\mathrm{SD}(q,u) \ =\ \tfrac{1}{2}\|q-u\|_1 \ \le\ \tfrac{1}{2}\sqrt{|\Omega|}\,\|q-u\|_2 \ =\ \tfrac{1}{2}\sqrt{|\Omega|\textstyle\sum_\omega q_\omega^2 - 1}\,. \tag{3}$$

Proof. For (2), $1 = (\sum_{\omega\in A} q_\omega)^2 \le |A|\sum_{\omega\in A} q_\omega^2 \le |A|\sum_\omega q_\omega^2$. For (3), the inequality is $\|v\|_1 \le \sqrt{|\Omega|}\,\|v\|_2$ applied to $v = q-u$, and the final equality is $\|q-u\|_2^2 = \sum_\omega q_\omega^2 - 1/|\Omega|$, which follows by expanding the square and using $\sum_\omega q_\omega = 1$. □

2 The leftover-hash step

Proposition 1 (The bound the lemma returns). *For every table h and every view z of positive probability,*

$$\max_{\mathsf{D}}\big(2\Pr[\mathsf{D}\ \textit{wins} \mid h,z] - 1\big) \ \le\ \tfrac{1}{2}\sqrt{R\,\bar{c} - 1}\,,$$

and consequently, for every unbounded D, $\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D}) \le \mathbb{E}_{(H,Z)}\big[\tfrac{1}{2}\sqrt{R\,\bar{c}-1}\big]$.

Proof. Because sd is drawn only after S has terminated, it is uniform on $\mathcal{K}$ and independent of X given (h,z). The distinguisher's view therefore differs between the two challenge bits only in the pair (sd, y_b), which is distributed as $Q := \mathrm{law}(sd, h(sd,X))$ under $b = 0$ and as the uniform distribution U on $\mathcal{K}\times\mathcal{R}$ under $b = 1$. The optimal advantage of an unbounded distinguisher between two distributions is their statistical distance, so the left-hand side is $\mathrm{SD}(Q,U)$.

Now $|\mathcal{K}\times\mathcal{R}| = KR$ and

$$\sum_{k,r} Q(k,r)^2 \ =\ \Pr\big[(sd,h(sd,X)) = (sd',h(sd',X'))\big] \ =\ \sum_k \frac{1}{K^2}\,c_k \ =\ \frac{\bar{c}}{K},$$

the middle equality because sd, sd' are independent and uniform, so both equal a given k with probability $1/K^2$, and given $sd = sd' = k$ the outputs collide with probability c_k. Feeding this into (3) gives $\mathrm{SD}(Q,U) \le \frac{1}{2}\sqrt{KR\cdot\bar{c}/K - 1} = \frac{1}{2}\sqrt{R\bar{c}-1}$, the radicand being nonnegative by (2) applied to Q with $A = \mathcal{K}\times\mathcal{R}$. Averaging over (H,Z) gives the second claim. □

Writing $c_k = \mathrm{CP} + \beta_k$ with $\beta_k := \sum_{x \neq x'} p_x p_{x'} \mathbf{1}[h(k,x) = h(k,x')]$ and $\bar{\beta} := \frac{1}{K} \sum_k \beta_k$ splits the quantity under the root as

$$R\bar{c} - 1 \ = \ R \cdot \mathrm{CP} \ + \ \gamma, \qquad \gamma \ := \ R\bar{\beta} - 1, \tag{4}$$

the first summand carrying the source's entropy deficiency and the second the family's failure to be universal on the source's support. Which of the two a lower bound on $R\bar{c} - 1$ speaks about is exactly the question Lemma 1 settles, and it must be settled: a bound on a sum is not by itself a bound on either part.

3 Calibration: on a universal table the lemma is sharp

Proposition 2 (Universal families). *Fix a table h and suppose the family $\{h(k,\cdot)\}_{k \in \mathcal{K}}$ is 2-universal, that is $\Pr_k[h(k,x) = h(k,x')] \leq 1/R$ for all $x \neq x'$. Then $\gamma \leq -\mathrm{CP} \leq 0$ for every source law p, and the conditional bound of Proposition 1 is at most $\frac{1}{2}\sqrt{R\,\epsilon_{h,z}}$. Consequently, if* every *table in the support of H has this property, then $\mathbf{Adv}^{\text{ext-pub}}(S, \mathsf{D}) \leq \frac{1}{2}\sqrt{\epsilon R}$ for every ϵ-unpredictable S and every unbounded* D.

Proof. Exchanging the order of summation, $\bar{\beta} = \sum_{x \neq x'} p_x p_{x'} \Pr_k[h(k,x) = h(k,x')] \leq (1 - \mathrm{CP})/R$, valid for every p because the hypothesis is a statement about every pair $x \neq x'$; so $\gamma \leq 1 - \mathrm{CP} - 1 = -\mathrm{CP}$. By (4) the conditional bound is at most $\frac{1}{2}\sqrt{R \cdot \mathrm{CP} - \mathrm{CP}} \leq \frac{1}{2}\sqrt{R\,\epsilon_{h,z}}$, using $\mathrm{CP} \leq \epsilon_{h,z}$. For the last claim, average over (H, Z) and apply Jensen's inequality to the concave nondecreasing $\sqrt{\cdot}$ — legitimate although p depends on H, since the conditional bound holds pointwise in (h, z) — together with $\mathbb{E}[\epsilon_{H,Z}] \leq \epsilon$. □

The hypothesis is a property of a single table, whereas H in Section 1 is uniform on all of $\mathsf{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$; the second sentence of Proposition 2 is therefore about a different and more favourable model than the one at issue. It is stated to calibrate, not to apply. What it calibrates is this: on the first summand of (1) the leftover hash lemma gives $\frac{1}{2}\sqrt{\epsilon R}$, which is that summand with $c = \frac{1}{2}$. Nothing is lost there, and everything that separates the two routes therefore lies in γ.

4 The limitation

Proposition 2 does not apply here. Universality is not merely absent from a typical table; at the seed lengths at issue it is unavailable from any table, provided the output is shorter than the input.

Remark 1 (No universal family at small K, when $R < D$). If $\{h(k,\cdot)\}_{k \in \mathcal{K}}$ is 2-universal then, viewing each input x as the column vector $(h(k,x))_{k \in \mathcal{K}} \in \mathcal{R}^K$, two distinct columns agree in at most K/R coordinates, so the D columns form a code of length K over an alphabet of size R with minimum distance at least $(1 - 1/R)K$. The Plotkin bound, in the form "$d = (1 - 1/q)n$ implies $|C| \leq qn$", gives $D \leq RK$, that is $K \geq D/R$. So no 2-universal family exists once $K < D/R$, which at $D = 2^{256}$ and $R = 2^{64}$ holds for every $K < 2^{192}$.

The proviso matters. For $R \geq D$ a single injective row is already 2-universal, with $K = 1$, so no seed-length lower bound holds without a hypothesis relating R to D. An earlier version of this note asserted $K \geq D^{\Omega(1)}$ unconditionally, which that example refutes.

So γ must be taken over the supports the source may choose after reading H, and the theorem below says how large those make $R\bar{c} - 1$. Recall that the source sees the entire table before *sd* exists, so any support it can name from H is admissible.

Theorem 1 (The collision route cannot beat $\sqrt{W/K}$). *Let $\mathcal{K}, \mathcal{D}, \mathcal{R}$ be nonempty and finite and let $\epsilon \in [1/D, 1]$. Put*

$$t := \lceil 1/\epsilon \rceil, \qquad \sigma := \lceil Rt/D \rceil, \qquad W := \lfloor R/\sigma \rfloor.$$

Then $1 \leq t \leq D$, $1 \leq \sigma \leq R$ and $1 \leq W \leq R$, and for every *table $h \in \mathsf{Fun}(\mathcal{K} \times \mathcal{D}, \mathcal{R})$ there is a source S — uniform on a set of t inputs, with $z = \perp$, and ϵ-unpredictable — for which*

$$R\,\bar{c} - 1 \;\geq\; \frac{W-1}{K}, \qquad \textit{so that the bound of Proposition 1 is at least} \qquad \frac{1}{2}\sqrt{\frac{W-1}{K}}.$$

No randomness is used: this holds for every table, not merely for most. When R and D/t are powers of two, as in the parameter ranges of interest, $W = \min\{R,\ D/t\}$ exactly; in general $W > \frac{1}{2}\min\{R,\ D/t\} - 1$.

Proof. The parameters are in range. From $\epsilon \leq 1$ we get $t \geq 1$, and from $\epsilon \geq 1/D$ that $t = \lceil 1/\epsilon \rceil \leq D$; hence $Rt/D \leq R$ and, R being an integer, $\sigma = \lceil Rt/D \rceil \leq R$, while $\sigma \geq 1$ because $Rt/D > 0$. From $\sigma \leq R$ we get $W = \lfloor R/\sigma \rfloor \geq 1$, and from $\sigma \geq 1$ that $W \leq R$.

The support. Fix any seed $k_1 \in \mathcal{K}$ and let $\mathcal{S} \subseteq \mathcal{R}$ consist of σ symbols r maximising $|\{x \in \mathcal{D} : h(k_1, x) = r\}|$. The D inputs are distributed among the R symbols, so the σ most popular of them carry at least a σ/R fraction of the total, that is at least $D\sigma/R$ inputs; and $\sigma \geq Rt/D$ gives $D\sigma/R \geq t$. So $\{x : h(k_1, x) \in \mathcal{S}\}$ has at least t elements; let T be any t of them and let $S(h)$ output $x \xleftarrow{\$} T$ together with $z \leftarrow \perp$. This is a counting fact about an arbitrary map $\mathcal{D} \to \mathcal{R}$; no property of h beyond being a function is used.

Admissibility. T is computed from h, which S receives in full, and *sd* is drawn only after S terminates, so nothing in the construction depends on the seed and S is a legitimate source for the game. Its conditional law is uniform on T, so $\epsilon_{h,\perp} = 1/t = 1/\lceil 1/\epsilon \rceil \leq \epsilon$; since this holds for every table, $\mathbb{E}[\epsilon_{H,Z}] \leq \epsilon$ and S is ϵ-unpredictable in the sense of Section 1.

The two collision bounds. For every k the law of $h(k, X)$ is a distribution on $\mathcal{R}$, so (2) with $A = \mathcal{R}$ gives $c_k \geq 1/R$. For $k = k_1$ every $x \in T$ has $h(k_1, x) \in \mathcal{S}$, so the law of $h(k_1, X)$ is carried by $\mathcal{S}$ and (2) with $A = \mathcal{S}$ gives $c_{k_1} \geq 1/\sigma$.

Assembling. Averaging the K lower bounds and using $R/\sigma \geq \lfloor R/\sigma \rfloor = W$,

$$R\,\bar{c} - 1 \;\geq\; R\Big[\frac{1}{K}\cdot\frac{1}{\sigma} + \frac{K-1}{K}\cdot\frac{1}{R}\Big] - 1 \;=\; \frac{R/\sigma}{K} + \frac{K-1}{K} - 1 \;\geq\; \frac{W}{K} - \frac{1}{K} \;=\; \frac{W-1}{K},$$

and Proposition 1 applied to this source returns $\frac{1}{2}\sqrt{R\bar{c} - 1} \geq \frac{1}{2}\sqrt{(W-1)/K}$.

The two forms of W. If $R = 2^m$ and $D/t = 2^\ell$ then $Rt/D = 2^{m-\ell}$, so $\sigma = \max\{1, 2^{m-\ell}\}$ and $W = \lfloor R/\sigma \rfloor = \min\{2^m, 2^\ell\} = \min\{R, D/t\}$. In general $\sigma < Rt/D + 1$, so, writing $a := R$ and $b := D/t$,

$$\frac{R}{\sigma} > \frac{a}{a/b+1} = \frac{ab}{a+b} = \frac{\min\{a,b\}\cdot\max\{a,b\}}{a+b} \geq \tfrac{1}{2}\min\{a,b\},$$

the last step because $a+b \leq 2\max\{a,b\}$; hence $W > R/\sigma - 1 > \frac{1}{2}\min\{R, D/t\} - 1$. □

Remark 2 (Pinning more than one row). Confining j rows rather than one, each to σ symbols, gives $R\bar{c} - 1 \geq j(W-1)/K$ by the same averaging, provided a common support of size t survives. It does: choosing the j symbol sets $\mathcal{S}_1, \ldots, \mathcal{S}_j$ independently and uniformly among the subsets of $\mathcal{R}$ of size R/σ, each input survives all j constraints with probability σ^{-j}, so the expected number of survivors is $D\sigma^{-j}$ and some choice attains it; the constraint is $D\sigma^{-j} \geq t$, that is $j \leq \log_\sigma(D/t)$. The gain is a factor $\sqrt{j}$ in the bound, hence $\log_2 j$ further bits of seed in the comparison below — one or two in the parameter ranges of interest — so the one-row construction of Theorem 1 is essentially the whole effect.

The averaging is the point. Choosing each $\mathcal{S}_i$ greedily, as the σ commonest symbols of its row, does *not* work for a worst-case table, since the j greedy sets may have a small or even empty common preimage; an earlier version of this note asserted the greedy version. Only the one-row case survives greedily, and that is the case Theorem 1 uses.

5 When the floor is a statement about γ

Theorem 1 bounds $R\bar{c} - 1 = R \cdot \mathrm{CP} + \gamma$ from below, and that is a bound on the sum. It carries no information about γ unless the floor exceeds what the first summand supplies on its own, which for the constructed source is exactly $R \cdot \mathrm{CP} = R/t$.

Lemma 1 (Attribution). *For the source of Theorem 1,* $\mathrm{CP} = 1/t$ *exactly, so* $\gamma \geq (W-1)/K - R/t$. *In particular, if*

$$\frac{W-1}{K} > \frac{R}{t}, \tag{5}$$

then the floor of Theorem 1 is not explained by the entropy deficiency, and $\gamma > 0$*: the family is genuinely non-universal on that support, by a margin of at least* $(W-1)/K - R/t$.

Proof. The source is uniform on T with $|T| = t$, so $\mathrm{CP} = \sum_{x\in T} t^{-2} = 1/t$. Substituting into (4) and rearranging Theorem 1's conclusion gives $\gamma = R\bar{c} - 1 - R/t \geq (W-1)/K - R/t$, and (5) makes it positive. □

Condition (5) is not automatic, and the comparison below is stated only where it holds. Where it fails — for instance $R = D$ and $\epsilon = 1/2$, so $t = 2$, $\sigma = 2$, $W = \lfloor R/2 \rfloor$, and $R/t = R/2$ exceeds $(W-1)/K$ for every $K \geq 1$ — the theorem is consistent with $\gamma \leq 0$, that is with a universal family, and says nothing about the second summand.

Corollary 1 (The comparison, in seed length). *Fix a target error* $\delta \in (0,1)$ *and parameters for which (5) holds and the entropy deficiency is not itself binding, that is* $c\sqrt{\epsilon R} < \delta$.

Driving the second summand of (1) *below* δ *requires* $K \geq c^2 \log_2 D/\delta^2$*, whereas by Theorem 1 the leftover-hash route cannot certify* δ *at all unless* $K \geq (W-1)/(4\delta^2)$*. In bits,*

$$\begin{aligned} \log_2 K &= 2\log_2 c + \log_2 \log_2 D + 2\log_2(1/\delta) && \textit{(conjectured)}, \\ \log_2 K &\geq \log_2(W-1) - 2 + 2\log_2(1/\delta) && \textit{(collision route)}. \end{aligned}$$

Both carry $2\log_2(1/\delta)$*; they differ in the first term,* $\log_2 W$ *against* $\log_2 \log_2 D$*.*

Side by side, and this is the whole comparison in one line:

$$\underbrace{c\left(\sqrt{\epsilon R} + \sqrt{\frac{\log_2 D}{K}}\right)}_{\text{conjectured, and proved in } \texttt{proof.tex}} \qquad \text{against} \qquad \underbrace{\frac{1}{2}\sqrt{\epsilon R + \frac{W-1}{K}}}_{\text{the least the collision route can return}} \quad . \tag{6}$$

Since $\sqrt{a+b} \geq (\sqrt{a}+\sqrt{b})/\sqrt{2}$ by the arithmetic–geometric mean inequality, the right-hand side is at least $\frac{1}{2\sqrt{2}}(\sqrt{\epsilon R} + \sqrt{(W-1)/K})$, so the two are the same two ingredients up to constants and differ in exactly one place: what stands under the second square root. The conjecture pays $\log_2 D$, a number of bits; the collision route pays W, the size of a set.

Remark 3 (A worked instance). Take $D = 2^{256}$, $R = 2^{64}$, a source of min-entropy 160 (so $\epsilon = 2^{-160}$ and $t = 2^{160}$), $\delta = 2^{-32}$ and $c = 2$. Then $\sigma = \lceil 2^{-32} \rceil = 1$ and $W = R = 2^{64}$. Both hypotheses of Corollary 1 hold: the deficiency term is $c\sqrt{\epsilon R} = 2^{-47} \ll \delta$; and at the seed length (1) requires, $K = 2^{74}$, the floor is $(W-1)/K \approx 2^{-10}$ against $R/t = 2^{-96}$, so (5) holds with room to spare and Lemma 1 gives $\gamma > 0$. The comparison is then $\log_2 K = 74$ under (1) against $\log_2 K \geq 126$ for the collision route: a gap of 52 bits.

An earlier version of this note used min-entropy 128 here, where $c\sqrt{\epsilon R} = 2^{-31}$ exceeds $\delta = 2^{-32}$, so (1) could not certify that δ at any seed length and the comparison was empty. The hypotheses of Corollary 1 are not decorative.

Remark 4 (Where the loss comes from, and when it vanishes). The loss is the step (3), $\|v\|_1 \leq \sqrt{KR}\,\|v\|_2$, which is tight only when v is spread evenly over all KR cells; the source of Theorem 1 makes it far from that, its deviation on the pinned row being a single concentrated lump.

The true advantage against that source should not be mis-stated, and an earlier version of this note did mis-state it as $O(1/K)$. Since $\mathrm{SD}(Q, U) = \frac{1}{K}\sum_k \mathrm{SD}(\mathrm{law}\, h(k, X), U_{\mathcal{R}})$, the pinned row k_1 contributes about $1/K$, but the other $K-1$ rows are *not* uniform. On a typical table each is t draws into R bins and contributes $\Theta(\sqrt{R/t})$, so the true advantage is $\Theta(\sqrt{R/t}) + \Theta(1/K)$, dominated by the *unpinned* rows in the regime of Remark 3; and on an adversarial table — every row constant, say — it is close to 1. So the overshoot of the lemma over the truth is neither $\sqrt{RK}$ nor uniform in the parameters. In the instance of Remark 3, at the seed length (1) requires, the lemma returns $\frac{1}{2}\sqrt{(W-1)/K} = 2^{-6}$ against a true advantage of about $0.4\sqrt{R/t} = 2^{-49.3}$; at the much larger seed length the lemma itself requires, the two come within a small factor of one another. The floor is a statement about what the technique can certify, not about how weak the source is.

The gap closes when W does. At $\epsilon = 1/D$ one has $t = D$, $\sigma = R$, $W = 1$ and Theorem 1 is trivial, correctly: a source of maximal entropy is uniform on all of $\mathcal{D}$ and has no support to select. More generally W is small whenever the source's min-entropy is close to $\log_2 D$ or the output is short: for $D = 2^{256}$, $R = 2^{64}$ and min-entropy 254 one has $t = 2^{254}$, $\sigma = 2^{62}$, $W = 4$, and the theorem exhibits no loss at all.

6 Reading the difference as key derivation

The game is a key-derivation problem. The input x is a secret with entropy but no uniformity — a Diffie–Hellman element, a biometric reading, the output of a physical source; sd is a public salt; H is a public hash; and $H(sd, x)$ is the derived key, which must look uniform to an adversary who knows H in full and sees sd. In that reading $\log_2 K$ is the *salt length* in bits, $m := \log_2 R$ the derived key length, $n := \log_2 D$ the length of the secret's encoding, and $\log_2(1/\epsilon)$ its min-entropy.

Whenever the derived key is no longer than the source's entropy gap, that is $m \le n - \log_2(1/\epsilon)$, one has $W = R$ and the two salt lengths of Corollary 1 read

$$\underbrace{\log_2 n + 2\log_2(1/\delta) + 2}_{\text{conjectured}} \qquad \text{against} \qquad \underbrace{m + 2\log_2(1/\delta) - 2}_{\text{collision route}}.$$

They differ in the first term, and they differ in kind. The conjectured salt grows with the *logarithm of the secret's length*; the collision route's salt grows with *the length of the key being derived*. So under the conjecture, deriving a 256-bit key rather than a 128-bit one from the same secret costs no extra salt whatever, while under the collision route it costs 128 further bits of salt. Concretely, with $c = 2$ and in each row the hypotheses of Corollary 1 verified:

	parameters				salt needed (bits)			
	$\log_2 D$	min-ent.	key	δ	conj.	route	univ.	RO†
DH element, 2048-bit group	2048	256	128	2^{-32}	77	190	1920	0
the same, 256-bit key	2048	512	256	2^{-64}	141	382	1792	0
a 512-bit source	512	300	128	2^{-64}	139	254	384	0

† The unsalted random-oracle heuristic of [1], Remark 5: $\epsilon_{\mathrm{RO}} \approx T/2^{\mu}$, so no salt is needed at all. It is in the table because that zero is the most informative entry in the row, and because it is bought rather than free. The bound holds only against a distinguisher of running time $T \le \delta\, 2^{\mu}$ — that is $T \le 2^{224}$, 2^{448} and 2^{236} in the three rows — and only for a source independent of H. Those ceilings are enormous, which is why the heuristic is attractive in practice; but they are ceilings, and the whole of `main.tex` lives above them, where every party reads the entire table and $T/2^{\mu}$ says nothing.

The last column is universal hashing, at $n - m$ bits from $K \ge D/R$ (Remark 1), and it is included because it makes the point of Section 7 concrete: universal hashing does not have a shorter salt than the conjecture, it has a far longer one. It is also read slightly differently from the other two. Those are the salt needed to reach the stated δ; this is the salt needed for such a family to *exist* at all, the binding constraint being existence rather than accuracy — at that salt the error is $\frac{1}{2}\sqrt{\epsilon R}$, which is 2^{-65}, 2^{-129} and 2^{-87} in the three rows, already inside δ in each.

The three columns scale in three different ways, and that is the whole comparison in one place:

$$\underbrace{\log_2 n + 2\log_2(1/\delta) + 2,}_{\text{conjectured}} \qquad \underbrace{m + 2\log_2(1/\delta) - 2,}_{\text{collision route}} \qquad \underbrace{n - m}_{\text{universal}}\ .$$

The conjectured salt grows with the logarithm of the secret's length; the collision route's grows with the length of the derived key; universal hashing's grows with the secret's length outright, *shrinks* as the derived key gets longer — 1920 bits for a 128-bit key against 1792 for a 256-bit one, since a larger R weakens the constraint $K \geq D/R$ — and does not depend on δ at all, because it is a structural requirement rather than an accuracy one. Only the first is what one wants of a salt: short, and growing only in the logarithm of anything.

Read the other way — at a fixed salt rather than a fixed error — the same gap says the route certifies almost nothing at the salt length the conjecture licenses. Substituting $K = c^2 \log_2 D/\delta^2$ into Theorem 1's floor gives

$$\frac{1}{2}\sqrt{\frac{W-1}{K}} \approx \frac{\delta}{2c}\sqrt{\frac{W}{\log_2 D}},$$

so the error the route can certify exceeds δ by a factor of about $\sqrt{W/\log_2 D}\,/\,(2c)$ — roughly $(m - \log_2 n)/2$ bits of security, over half the derived key's length. In the first row of the table that factor is $2^{56.5}$, so at a 77-bit salt the route certifies $2^{24.5}$: larger than 1, hence nothing at all. In the third row it certifies $2^{-6.5}$ at a 139-bit salt — not vacuous, but six bits of security where sixty-four were wanted.

Neither reading says that salting with 77 bits is unsafe. Both say that the collision route cannot be what establishes it, and Remark 4 is the reminder that the true advantage against the witness source is far below either bound. What is being measured here is the reach of a proof technique, not the security of a construction.

7 Is the random oracle worse than universal hashing?

Proposition 2 gives a universal family the bound $\frac{1}{2}\sqrt{\epsilon R}$ with no second summand at all, while Theorem 1 saddles a random table with a floor of $\frac{1}{2}\sqrt{(W-1)/K}$ on top of it. Read quickly, that says the random oracle is the worse extractor, which would be a strange thing for a random function to be. It is worth saying exactly why the reading is wrong, because the error is a natural one.

The two are not being compared at the same seed length. By Remark 1 a 2-universal family on D inputs needs $K \geq D/R$, a salt of $n - m$ bits. That is what buys $\gamma \leq 0$. The random-oracle question is posed at $K \approx c^2 \log_2 D/\delta^2$, a salt of $\log_2 n + 2\log_2(1/\delta)$

bits, which is smaller by an exponential factor. Comparing the two bounds without comparing their seeds is comparing the price of two objects while looking only at one price tag.

At equal seed length the random oracle matches, and the gap closes completely. Give the random table the salt a universal family requires, $K = D/R$. Then the conjectured second summand is

$$\sqrt{\frac{\log_2 D}{K}} = \sqrt{\frac{R \log_2 D}{D}},$$

which for $D = 2^{2048}$ and $R = 2^{128}$ is $2^{-954.5}$: not small, but absent. Both routes then return $\Theta(\sqrt{\epsilon R})$ and there is nothing to choose between them. The second summand is not a defect the random oracle carries and universal hashing escapes; it is the price of a *short* salt, and universal hashing never pays it only because it never has one.

At the seed length that makes the problem interesting, universal hashing does not exist. At the parameters of Section 6's first row — a Diffie–Hellman element in a 2048-bit group, min-entropy 256, a 128-bit key, $\delta = 2^{-32}$ — the conjectured bound does the job with a 77-bit salt. A universal family needs at least 1920 bits, twenty-five times as long, to do the same job no better: its error there is $\frac{1}{2}\sqrt{\epsilon R} = 2^{-65}$ against the random oracle's 2^{-63} at the same salt. Measured the way extractors are normally measured — error achieved per bit of seed — universal hashing is the seed-suboptimal construction and the random oracle is the optimal one. This is the standard picture, not a novelty of this setting: the leftover hash lemma is known to be far from optimal in seed length, and (1) sits at the information-theoretic optimum, which universal hashing misses by an exponential margin.

Why universality is immune to support selection is why it is expensive. This is the part worth keeping. The reason Proposition 2 survives a source that reads the whole table is visible in its proof: the bound $\bar{\beta} = \sum_{x \neq x'} p_x p_{x'} \Pr_k[h(k,x) = h(k,x')] \leq (1 - \mathrm{CP})/R$ holds for *every* law p, because universality constrains *every* pair $x \neq x'$ separately. A worst-case-over-all-pairs hypothesis is automatically robust to an adversary who picks the pairs last. But a worst-case-over-all-pairs hypothesis is exactly what Remark 1's counting argument charges D/R seeds for. The immunity and the cost are the same property seen twice. A random table satisfies only the average-case version, which is why choosing the support after seeing H can hurt it at all — and why $\log_2 n$ bits of salt suffice to stop that, rather than $n - m$.

And the floor is about the technique, not the object. Theorem 1 bounds what one particular accounting can certify about a random table. It is not a lower bound on the advantage, and the random table is in fact better than the floor suggests: (1) is *true*, proved in `proof.tex` with $c = 2$, so the random oracle does achieve $\sqrt{\epsilon R} + \sqrt{\log_2 D/K}$ at a short salt. What Theorem 1 says is that the collision route cannot be the proof of it. Remark 4 makes the same point numerically: against the witness source the true advantage is some 2^{-49} where the route returns 2^{-6}.

8 Prior art: the bound is Stinson's

Theorem 1 is not new, and this section says so precisely, because the rest of the note would otherwise read as though it were.

Barak, Dodis, Krawczyk, Pereira, Pietrzak, Standaert and Yu [1] open by naming two limitations of the leftover hash lemma. The first is entropy loss. The second is the one at issue here: for an (almost) universal family good enough for the LHL, the seed length must be at least

$$\min\bigl(u - v,\ v + 2\log(1/\varepsilon)\bigr) - O(1),$$

where u is the length of the source and v the number of extracted bits, "and, thus, must grow with the number of extracted bits". They attribute it to Stinson [2], and record separately that Stinson showed perfectly universal families need a seed linear in u.

Translating — $u = \log_2 D$, $v = \log_2 R = m$, $\varepsilon = \delta$, and the seed length is $\log_2 K$ — that bound reads $\log_2 K \geq \min(\log_2 D - m,\ m + 2\log_2(1/\delta)) - O(1)$. Corollary 1 gives $\log_2 K \geq \log_2(W-1) - 2 + 2\log_2(1/\delta)$, and in the principal regime $W = R$ that is $m + 2\log_2(1/\delta) - 2$: the second branch. The first branch is Remark 1's $K \geq D/R$, which is Stinson's linear-in-u result reached by another route. So the two salt columns of Section 6's table are the two branches of one bound that has been known since 1994, and Section 7's reading — that universal hashing has the longer salt, not the shorter — is that bound's own reading. What this note adds is not the number.

What it does add is three things, none of them a new bound. First, a self-contained derivation of the relevant branch from a pigeonhole and two applications of Cauchy–Schwarz, with no probability in it, so the statement holds for every table rather than a typical one. Second, that derivation instantiated in the game of `main.tex`: Theorem 1 exhibits a witness source per table, so the obstruction appears as an attack one can write down rather than as a counting bound on families. Third, Lemma 1, which separates the cases where the floor speaks about γ from those where the entropy deficiency already explains it. Whether any of the three is also in Stinson's papers I do not know: the attribution above is [1]'s, and I have not read [2] at first hand.

Two things in the game of `main.tex` that literature does *not* cover, and they are why the question is not simply closed by it.

- **The independence hypothesis is *not* weakened here, and it is worth being exact about that, because the natural summary is wrong.** Definition 2.1 of [1] quantifies over all distributions X with $\tilde{H}_\infty(X \mid Z) \geq m$, with the seed S drawn independently as the coins of a function Ext that is fixed beforehand and public. So a source there may already depend on the *family*; what it must not depend on is the *seed*. The game of `main.tex` asks no more than that: *sd* is drawn after S has terminated, so its source is independent of the seed too. Nothing in the adversary model is weakened, and Theorem 1's witness uses no freedom the leftover-hash setting withholds. What differs is the *family* — a uniformly random table in place of a universal one — and the content of Theorem 1 is that this substitution alone, with seed-independence fully intact, already costs the seed length. (That paper's remark about not making "the source X dependent on the seed" is about a different scenario, fixing one public seed and reusing it indefinitely, and does not bear on this.)

- **Their random-oracle comparison bounds one party only, and it is unsalted.** This is set out in full in Remark 5 below, because the asymmetry is the whole of why that comparison does not reach this setting.

Remark 5 (The query model of the random-oracle comparison). Section 3.4 of [1] compares the leftover hash lemma against the "dream" bound obtained by modelling a cryptographic hash as a random oracle. Its construction is *unsalted*: the derived key is $H(X)$, with no seed anywhere in the argument. The reasoning is that $H(X)$ is distinguishable from uniform only if the attacker queries the oracle at X; as X has min-entropy μ, that happens with probability at most $T/2^{\mu}$, where T bounds the adversary's running time. Calibrating T against the application's own security — a primitive P that is ϵ-secure against complexity T must satisfy $\epsilon \leq T/2^{v}$, since exhaustive search over a v-bit key must not succeed better than ϵ, so $T \approx \epsilon 2^{v}$ — gives

$$\epsilon_{\mathrm{RO}} \approx \frac{T}{2^{\mu}} = \epsilon\, 2^{v-\mu} = \epsilon\, 2^{-L}, \qquad L := \mu - v \text{ the entropy loss,}$$

against $\epsilon 2^{-L/2}$ for their generalised LHL and $2^{-L/2}$ for the standard one.

Only one of the two parties is bounded, and the two are not bounded in the same sense. The *distinguisher* is bounded, by the running time T, which in particular bounds its oracle queries. The *source* is not bounded — because it is not a query-making object at all. There X is a distribution of min-entropy μ, fixed independently of H; no source algorithm with oracle access appears in the model, so there is nothing about it to bound. Searching [1] for query counts finds only the distinguisher's T and the query budgets of the *applications* being keyed (encryption, MAC and PRF queries); the source never makes a query.

That asymmetry is exactly what the argument rests on, and it is not incidental. "$H(X)$ is distinguishable only if the attacker queried at X" is false the moment X may depend on H: a source that reads H and returns an x whose image has a fixed first bit is distinguished with no query at all. So this comparison needs the source independent of H *itself* — a strictly stronger hypothesis than the leftover-hash half of the same paper needs, where the source must be independent only of the seed and may know the family. Being unsalted, it has no seed to fall back on, so independence from H is the only thing holding it up. That is the hypothesis `main.tex` does drop, and the salt is what it puts in its place.

There is also no formal random-oracle model to appeal to here: [1]'s Section 3.4 is prose — "it is instructive to compare…the heuristic 'dream' bound…we claim" — and the random oracle appears in no definition, lemma or theorem of that paper. Its formal Definition 2.1 has Ext a plain function and X a distribution quantified over, with no oracle anywhere. So the question of how many oracle queries the source may make has no answer there, not because the answer is zero but because the object is a distribution rather than an algorithm.

It is also why that comparison needs no salt while `main.tex` does. Once the distinguisher is unbounded the estimate $T/2^{\mu}$ is vacuous: in the game of `main.tex` every party holds all KD entries of the table, so T is effectively KD and $T/2^{\mu} = KD\epsilon$ exceeds 1 for every interesting ϵ. Unsalted extraction against an unbounded distinguisher simply fails, and this note's own arithmetic agrees: at $K = 1$ the bound (1) reads $c(\sqrt{\epsilon R} + \sqrt{\log_2 D})$, which is vacuous for every $D \geq 2$. The salt is what rescues it, and the salt is what the question is about. Summarising the two models:

	[1], Section 3.4	`main.tex`
construction	$H(X)$, unsalted	$H(sd, x)$, salted
source	a distribution, independent of H; makes no queries	an algorithm reading all of H; unbounded
distinguisher	running time $\leq T$	unbounded; holds all of H
bound	$\epsilon\, 2^{-L}$	$c(\sqrt{\epsilon R} + \sqrt{\log_2 D/K})$

9 The query-bounded model

Remark 5 sets two models side by side but examines only one. This section examines the other: the game of Section 1 with each party given oracle access in place of the table. Formally S^H makes at most q_S queries before returning (x, z); the seed $sd \xleftarrow{\$} \mathcal{K}$ is drawn afterwards, as before; and $\mathsf{D}^H(sd, y_b, z)$ makes at most q_D queries. Nothing else changes. In particular the *predictor* is left unbounded and still receives the whole table, so ϵ-unpredictability is the same hypothesis as in Section 1, the most demanding of the three, and every bound below is proved under it.

Two features of a source govern what follows, and neither is q_S. Writing $Q_S \subseteq \mathcal{K} \times \mathcal{D}$ for the set of cells the source queries, put

$$r_S \;:=\; \sum_{k \in \mathcal{K}} \Pr\big[Q_S \cap (\{k\} \times \mathcal{D}) \neq \emptyset\big], \qquad s_S \;:=\; \sum_{k \in \mathcal{K}} \Pr\big[(k, X) \in Q_S\big],$$

the expected number of *rows* the source touches and the expected number of seeds at which it queries *its own output*. Both are at most $\min(q_S, K)$ and both can be far smaller: the source of Theorem 1 reads an entire row, so $q_S = D$, yet $r_S = s_S = 1$.

What the collision route gives

Lemma 2 (An untouched row stays fresh). *Fix $k \in \mathcal{K}$, let A_k be the event that the source queries some cell of row k, and write $p_x^- := \Pr[X = x \wedge \neg A_k \mid H, Z]$. Then*

$$\mathbb{E}_{(H,Z)}\Big[\sum_{x \neq x'} p_x^-\, p_{x'}^-\, \mathbf{1}[H(k,x) = H(k,x')]\Big] \;\leq\; \frac{1}{R}.$$

Proof. Split the table as $H = (H_k, H_{-k})$, row k against the rest; the two are independent and H_k is uniform on $\mathsf{Fun}(\mathcal{D}, \mathcal{R})$. Take S deterministic given coins ρ, with ρ independent of H.

Run S on (ρ, h). Its transcript up to its first query into row k is a function of (ρ, h_{-k}) alone; hence so is the indicator of $\neg A_k$, and on $\neg A_k$ so is the whole transcript, and therefore (X, Z). Write

$$a(x,z,h_{-k}) := \Pr_{\rho}[X = x \wedge Z = z \wedge \neg A_k \mid h_{-k}], \qquad A(z,h_{-k}) := \sum_x a(x,z,h_{-k}),$$

so that $\Pr[H = h, Z = z, X = x, \neg A_k] = \Pr[h_{-k}]\Pr[h_k]\, a(x,z,h_{-k})$ and

$$\Pr[H = h, Z = z] \;=\; \Pr[h_{-k}]\Pr[h_k]\Pr_{\rho}[Z = z \mid h] \;\geq\; \Pr[h_{-k}]\Pr[h_k]\, A(z,h_{-k}).$$

Since $\Pr[H = h, Z = z]\, p_x^- = \Pr[H = h, Z = z, X = x, \neg A_k]$, the expectation in question is

$$\sum_{h,z} \frac{1}{\Pr[h,z]} \sum_{x \neq x'} \Pr[h,z,x,\neg A_k]\; \Pr[h,z,x',\neg A_k]\, \mathbf{1}[h(k,x) = h(k,x')],$$

and substituting the two displays above — the lower bound on $\Pr[h,z]$ in the denominator, which is the direction an upper bound needs — makes it at most

$$\sum_{h_{-k}} \Pr[h_{-k}] \sum_{h_k} \Pr[h_k] \sum_z \frac{1}{A(z,h_{-k})} \sum_{x \neq x'} a(x,z,h_{-k})\, a(x',z,h_{-k})\, \mathbf{1}[h_k(x) = h_k(x')].$$

Neither a nor A depends on h_k, so the sum over h_k may be performed first, and for $x \neq x'$ it gives $\sum_{h_k} \Pr[h_k]\mathbf{1}[h_k(x) = h_k(x')] = 1/R$ exactly. What remains is at most

$$\frac{1}{R} \sum_{h_{-k}} \Pr[h_{-k}] \sum_z \frac{A(z,h_{-k})^2}{A(z,h_{-k})} \;=\; \frac{1}{R} \sum_{h_{-k}} \Pr[h_{-k}]\, \Pr_{\rho}[\neg A_k \mid h_{-k}] \;\leq\; \frac{1}{R},$$

using $\sum_{x \neq x'} a\, a' \;\leq\; (\sum_x a)^2 \;=\; A^2$ and $\sum_z A(z,h_{-k}) \;=\; \Pr_\rho[\neg A_k \mid h_{-k}]$. Terms with $A(z,h_{-k}) = 0$ have vanishing numerator and are read as 0. □

Theorem 2 (The collision route, query-bounded). *For every source that is ϵ-unpredictable and every unbounded distinguisher,*

$$\mathbf{Adv}^{\text{ext-pub}}(S, \mathrm{D}) \;\leq\; \frac{1}{2}\sqrt{\epsilon R \;+\; \frac{2R\, r_S}{K}} \;\leq\; \frac{1}{2}\sqrt{\epsilon R \;+\; \frac{2R \min(q_S, K)}{K}}\;.$$

Proof. Fix (h,z) and k, write $u_k := \Pr[A_k \mid h,z]$ and split $p = p^+ + p^-$ into its parts on A_k and $\neg A_k$. Of the four terms of $c_k = \sum_{x,x'} p_x p_{x'} \mathbf{1}[h(k,x) = h(k,x')]$, the three meeting p^+ contribute at most $\sum_x p_x^+ \sum_{x'} p_{x'} + \sum_{x'} p_{x'}^+ = 2u_k$, the indicator being at most 1. Hence

$$c_k \;\leq\; 2u_k \;+\; \sum_x (p_x^-)^2 \;+\; \sum_{x \neq x'} p_x^-\, p_{x'}^-\, \mathbf{1}[h(k,x) = h(k,x')],$$

and $p^- \leq p$ pointwise makes the middle sum at most CP. Average over k and take the expectation over (H,Z). For the first term, $\mathbb{E}\big[\sum_k u_k\big] = \sum_k \Pr[A_k] = r_S$; for the second, $\mathbb{E}[\mathrm{CP}] \le \mathbb{E}[\epsilon_{H,Z}] \le \epsilon$; for the third, Lemma 2 gives at most $1/R$ for each of the K rows. So

$$\mathbb{E}[\,R\bar{c}-1\,] \;\le\; \frac{2R\,r_S}{K} + \epsilon R + R\cdot\frac{1}{R} - 1 \;=\; \epsilon R + \frac{2R\,r_S}{K}.$$

By Proposition 1 the advantage is at most $\mathbb{E}\big[\frac{1}{2}\sqrt{R\bar{c}-1}\big]$, and $\sqrt{\cdot}$ is concave, so Jensen's inequality moves the expectation inside the root. The radicand is nonnegative pointwise by Fact 1, so this is legitimate. □

Three readings of Theorem 2 are worth separating. It is a statement about *row breadth*, not query count: a source may spend D queries inside one row and remain capped at $\epsilon R + 2R/K$, while K queries placed one per row exhaust the bound. It degrades to nothing at $r_S = K$, correctly, since a source that pins every row leaves $\bar{c} = 1$. And at $r_S = 0$ it returns $\frac{1}{2}\sqrt{\epsilon R}$, the universal-family answer of Proposition 2: a source that never queries is independent of the table, and a random table is universal in expectation.

Proposition 3 (The dependence on r_S is exact up to a factor of two). *Fix $j \le K$ and let the source draw $x \xleftarrow{\$} \mathcal{D}$, query (k,x) for $k < j$, and output $\big(x, z = (H(0,x),\ldots,H(j-1,x))\big)$. It makes j queries and has $r_S = s_S = j$; it is ϵ-unpredictable with $\epsilon \le R^j/D$; and for* every *table*

$$R\bar{c} - 1 \;\ge\; \frac{(R-1)\,j}{K}.$$

Proof. Given (h,z) the conditional law is uniform on the joint fibre $T_z := \{x' : h(k,x') = z_k \text{ for all } k < j\}$, so $\epsilon_{h,z} = 1/|T_z|$, and since $\Pr[Z = z] = |T_z|/D$ the average $\mathbb{E}[\epsilon_{H,Z}]$ is the number of nonempty fibres divided by D, at most R^j/D. Every element of T_z has the same image in each pinned row, so $c_k = 1$ for $k < j$; for the remaining rows $c_k \ge 1/R$ by (2). Hence

$$R\bar{c} - 1 \;\ge\; R\Big[\frac{j}{K}\cdot 1 + \frac{K-j}{K}\cdot\frac{1}{R}\Big] - 1 \;=\; \frac{Rj}{K} - \frac{j}{K} \;=\; \frac{(R-1)j}{K}. \qquad \square$$

So the term $2R\,r_S/K$ of Theorem 2 cannot be improved below $(R-1)r_S/K$: the constant is off by at most a factor of two and the shape is right. At $j = K$ the proposition returns $R-1$, and the theorem is vacuous, as it must be.

Remark 6 (Theorem 2 closes the window of Section 10, for these sources). Theorem 1 is a floor and Section 10 names no matching ceiling; the one sketched there exceeds the floor by a factor of about 2^{21} in the instance of Remark 3. For a source of row breadth r_S the two now meet. In that instance — $D = 2^{256}$, $R = 2^{64}$, min-entropy 160, $K = 2^{74}$, where $W = R$ — the witness of Theorem 1 has $r_S = 1$, and

$$\frac{W-1}{K} \approx 2^{-10} \qquad \le \qquad R\bar{c}-1 \qquad \le \qquad \epsilon R + \frac{2R}{K} \;=\; 2^{-96} + 2^{-9} \;\approx\; 2^{-9}:$$

a window of a factor 2 in the radicand, so the route returns between 2^{-6} and $2^{-5.5}$. The hypotheses differ — Section 10's sketch asks the source to be uniform on a support and bounds no queries, Theorem 2 bounds r_S and asks nothing about the shape — but the witness satisfies both, and on it the query-bounded ceiling is the sharper by some twenty orders of magnitude.

Where q_D enters, and where it cannot

Remark 7 (No bound proved through Proposition 1 can mention q_D). Proposition 1 passes through $\mathrm{SD}(Q,U)$, and statistical distance *is* the supremum of the advantage over all distinguishers whatever. A bound obtained that way holds for the unbounded distinguisher and can therefore never improve as q_D falls. This is not a defect of the present derivation but of the route: to let the distinguisher's budget count one must leave the leftover hash lemma, and Theorem 3 is what one gets by doing so.

Theorem 3 (The bad-cell route). *For every ϵ-unpredictable source and every distinguisher making at most q_D queries,*

$$\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D}) \;\le\; \frac{s_S}{K} \;+\; q_D\,\epsilon \;\le\; \frac{\min(q_S,K)}{K} \;+\; q_D\,\epsilon.$$

Proof. Write the advantage as $|\Pr[b'=1 \mid b=0] - \Pr[b'=1 \mid b=1]|$ and couple the two worlds by sampling H lazily, with the same coins for S and D in both. They differ only in the challenge: the cell $H(sd,x)$ under $b=0$, an independent uniform value under $b=1$. Let BAD be the event that the cell (sd,x) is queried, by the source or by the distinguisher. Off BAD that cell is never read by anyone, so the two executions coincide step for step: they can first diverge only at a query to (sd,x), whence BAD occurs in one world exactly when it occurs in the other and the two views agree off it. The advantage is therefore at most $\Pr[\mathrm{BAD}]$, which we may compute in the $b=1$ world, and by a union bound at most $\Pr[(sd,x)\in Q_S] + \Pr[\mathsf{D}\text{ queries }(sd,x)]$.

For the first, sd is uniform on $\mathcal{K}$ and independent of the source's run, so $\Pr[(sd,X)\in Q_S] = \frac{1}{K}\sum_k \Pr[(k,X)\in Q_S] = s_S/K$, and $s_S \le \min(q_S,K)$ because the source can place at most q_S queries and there are only K rows.

For the second, build a predictor. Given (H,z), let P draw $sd \xleftarrow{\$} \mathcal{K}$ and $y \xleftarrow{\$} \mathcal{R}$, run $\mathsf{D}^H(sd,y,z)$ answering its queries from H, pick $i \xleftarrow{\$} \{1,\dots,q_D\}$ and return the input component of the i-th query. Under $b=1$ the challenge is a uniform value independent of everything else, so P's simulation is exact; and if D queries (sd,x) then some query has input component x, which P selects with probability at least $1/q_D$. So $\Pr[\mathsf{D}\text{ queries }(sd,x)] \le q_D\cdot \mathbf{Adv}^{\text{pred}}(P) \le q_D\epsilon$. □

Corollary 2 (Which route to use). *Under the hypotheses of both,*

$$\mathbf{Adv}^{\text{ext-pub}}(S,\mathsf{D}) \;\le\; \min\Big\{\, \frac{s_S}{K} + q_D\epsilon,\;\; \frac{1}{2}\sqrt{\epsilon R + \frac{2R\,r_S}{K}}\,\Big\},$$

and since $s_S/K \le \frac{1}{2}\sqrt{2R\,r_S/K}$ always, the two entropy terms decide between them: the bad-cell route is the smaller whenever

$$q_D \;\le\; \tfrac{1}{2}\sqrt{R/\epsilon}.$$

Proof. Both bounds are proved above; the comparison of entropy terms is $q_D\epsilon \le \frac{1}{2}\sqrt{\epsilon R}$, which is $q_D \le \frac{1}{2}\sqrt{R/\epsilon}$. For the aside: querying (k, X) touches row k, so $s_S \le r_S$; and $r_S/K \le \frac{1}{2}\sqrt{2Rr_S/K}$ is, on squaring, $r_S \le RK/2$, which holds because $r_S \le K$ and $R \ge 2$. □

That threshold is not a close call. In the instance of Remark 3 it is $q_D \le 2^{111}$; for a 128-bit derived key from a source of min-entropy 128 it is $q_D \le 2^{127}$. Below it — which is to say, in every parameter regime anyone would deploy — the leftover hash lemma is simply the wrong instrument for this game, and the gap is not marginal: on the witness of Theorem 1 in that instance, against a distinguisher making $q_D = 2^{80}$ queries, Theorem 3 returns $2^{-74} + 2^{-80} \approx 2^{-74}$ where Theorem 2 returns $2^{-5.5}$.

Remark 8 (What this settles about the witness). Remark 4 decomposes the true advantage against that witness as $\Theta(\sqrt{R/t}) + \Theta(1/K)$, the second summand from the pinned row and the first from the $K-1$ unpinned ones, and notes that the first dominates. Theorem 3 says where the first comes from: it is an artefact of the distinguisher being unbounded. Against a q_D-bounded distinguisher the unpinned rows contribute at most $q_D\epsilon = q_D/t$ in place of $\sqrt{R/t}$, which is smaller exactly when $q_D \le \sqrt{Rt} = \sqrt{R/\epsilon}$, the threshold of Corollary 2 up to its factor of two — while the pinned row still contributes its $s_S/K = 1/K$. So the reading that the advantage against this source is $\Theta(1/K)$ is correct precisely in the query-bounded model and false outside it, which is what Remark 4 found the hard way.

10 What is not claimed

Theorem 1 is one-sided. It says the collision route cannot return a bound better than $\frac{1}{2}\sqrt{(W-1)/K}$; it does *not* say the route attains that. Every statement in this note about the route's cost is therefore a necessary condition on K, never the cost itself, and Corollary 1 is phrased that way deliberately. Theorem 2 supplies a matching ceiling for sources of bounded row breadth, and Remark 6 closes the window to a factor of two on the witness itself; for sources that touch many rows the paragraph below still stands.

The remaining window is wide and should not be papered over. Certifying γ for a random table by a bounded-differences argument — $\bar{\beta}$ has bounded-differences constant $2/(Kt)$ in the Kt table entries above a support of size t, and a union bound over the $\binom{D}{t}$ supports of that size costs $e^{t(1+\ln D)}$ — yields, for sources uniform on a support, $\gamma \le R\sqrt{2(1+\ln D)/K}$ with high probability, hence an upper bound of $\frac{1}{2}\sqrt{\epsilon R} + \frac{1}{2}\sqrt{R}\,(2(1+\ln D)/K)^{1/4}$. Two gaps in that sketch are worth naming: it covers sources uniform on a support only, and a general source needs a further argument reducing to those; and the resulting ceiling exceeds Theorem 1's floor by a factor of order $\sqrt{R/W}\,(K\ln D)^{1/4}$, which in the instance of Remark 3 is about 2^{21}. So for a source of unrestricted row breadth the truth for this route lies somewhere in that window, and locating it is not attempted.

Nor is the bound one would most like in the query-bounded model of Section 9, namely $s_S/K + \frac{1}{2}\sqrt{\epsilon R}$: the pinned cells charged once each, and the leftover hash lemma applied to what is left as though it were universal. That would match the true advantage against the witness of Theorem 1 in both summands, and it is what one expects, since a row the source touched away from its own output is a row it has learnt nothing exploitable about. Proving it needs Lemma 2 refined from "row k untouched" to "the single cell (k, X) untouched", and that refinement is not available by the argument given: the freshness of one cell survives conditioning on the table only through the row-versus-rest factorisation the lemma uses, and two runs of the source may query in each other's output columns. Theorem 2 and Theorem 3 therefore stand as two separate bounds rather than one combined one, and Corollary 2 takes their minimum instead of their better parts.

What the floor alone does establish, in the regime where (5) holds and the deficiency term is not binding, is that the route cannot reach (1) there: it needs at least $\log_2(W-1) - 2\log_2 c - \log_2 \log_2 D - 2$ further bits of seed, which is 52 in the instance of Remark 3. Outside that regime nothing is claimed, and Remark 4 exhibits parameters — min-entropy 254 over the same D and R — where the construction shows no loss whatever.

Finally, the scope is one technique, not one lemma. Theorem 1 constrains the particular functional $\frac{1}{2}\sqrt{R\bar{c}-1}$ of Proposition 1, arrived at by bounding the collision probability of $(sd, h(sd, X))$ and passing to statistical distance through (3). It says nothing about other uses of the leftover hash lemma, and the honest form of the headline is that (1) is not a corollary of *that route*. Nothing here is a statement about (1) itself, which is true: `proof.tex` proves it with $c = 2$, by bounding the L_1 distance directly and never forming a collision probability at all.

[1] Boaz Barak, Yevgeniy Dodis, Hugo Krawczyk, Olivier Pereira, Krzysztof Pietrzak, François-Xavier Standaert and Yu Yu. Leftover Hash Lemma, Revisited. In *Advances in Cryptology — CRYPTO 2011*, LNCS 6841, pages 1–20. Springer, 2011. Cryptology ePrint Archive, Report 2011/088, https://eprint.iacr.org/2011/088. (The seed-length limitation quoted in Section 8 is the second bullet of its abstract and is developed on page 1 of the introduction; the query-bounded random-oracle comparison is its Section 3.4. Read at first hand.)

[2] Douglas R. Stinson. Universal hashing and authentication codes. *Designs, Codes and Cryptography*, 4(4):369–380, 1994. (Cited by [1] as the source of that seed-length bound. *Not* read at first hand for this note; the attribution is theirs.)

[3] Salil P. Vadhan. *Pseudorandomness.* Foundations and Trends in Theoretical Computer Science. Chapter 6, Randomness Extractors. (Theorem 6.17 is (1) in extractor parameters; the remark following Theorem 6.18 is the seed length of universal hashing. Read at first hand.)

[4] Jaikumar Radhakrishnan and Amnon Ta-Shma. Bounds for dispersers, extractors, and depth-two superconcentrators. *SIAM Journal on Discrete Mathematics*, 13(1):2–24, 2000. (The optimum $d \geq \log(n-k) + 2\log(1/\epsilon) - O(1)$ that (1) meets. Read at first hand.)